BANKSENSE ANALYTICS

Banking Customer & Transaction Analytics

End-to-End Financial Intelligence: Data Engineering, SQL Analytics, & Executive Power BI Dashboard

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STRATEGIC CONTEXT

Business Problem & Core Objectives

The Banking Challenge

Fragmented customer transaction logs hinder visibility into cash liquidity risks, customer retention drivers, and support bottlenecks across payment channels.

Project Objective

Build a robust relational data model and interactive Power BI dashboard to evaluate liquidity trends, segment high-value investors, and resolve support inefficiencies.

Target Business Value

Enable data-driven treasury cash-flow decisions, targeted VIP wealth management, and operational SLA optimization for customer care teams.

DATA FOUNDATION

Dataset & Banking Data Sources

Customers Table

200 unique retail banking client profiles with demographic & account tenure metadata.

Transactions Table

Granular logs of deposits ($2.06M total) and withdrawals ($900.93K total).

Bill Payments Table

Utility, credit, and service bill transactions dominating recurring user activity.

Support Tickets Table

400 customer service interactions tracking issue type and resolution status.

Investments Table

$3.39M in active portfolio assets across high-net-worth client accounts.

DATA ENGINEERING

Data Cleaning & Transformation (Power Query)

Handling Missing Values

Imputed missing categorizations and cleaned blank customer entries to maintain integrity.

Data Type Standardization

Cast currency fields to Fixed Decimal Number and formatted dates uniformly.

Dim_Date Creation

Generated a contiguous date dimension table with Year, Quarter, and Month attributes.

BACKEND VALIDATION

SQL Database & Exploratory Analysis

```sql
-- Executive Liquidity & VIP Segmentation Query
SELECT c.Customer_ID, c.Customer_Name,
  SUM(t.Deposit_Amount) AS Total_Deposits,
  SUM(i.Investment_Amount) AS Total_Investments
FROM Dim_Customers c
LEFT JOIN Fact_Transactions t ON c.Customer_ID = t.Customer_ID
LEFT JOIN Fact_Investments i ON c.Customer_ID = i.Customer_ID
GROUP BY c.Customer_ID, c.Customer_Name
ORDER BY Total_Investments DESC;
```

Relational Integrity

Validated primary and foreign key constraints across tables prior to Power BI loading.

Analytical Views

Built pre-aggregated SQL views for utility payment balances and ticket resolution rates.

PLATFORM ARCHITECTURE

Power BI Star Schema & Data Model

Star Schema Model

Central Date Dimension

Unified **Dim_Date** table connecting transactions, bill payments, and investments.

1:Many Relationships

Single-direction cross-filtering ensures exact multi-table aggregation without duplication.

Dynamic Slicers

Synchronized Year and Quarter dropdown controls driving real-time dashboard updates.

EXECUTIVE DASHBOARD OVERVIEW

Key Financial Performance Metrics

200
Total Customers

2.06M
Total Deposits ($)

900.93K
Total Withdrawals ($)

1.16M
Net Liquidity ($)

3.39M
Total Investments ($)

KEY INSIGHTS — PART 01

Bill Payment Behavior & Liquidity Dynamics

UTILITY BILLS VOLUME

$430,666.52

Dominates payment activity across 244 total transactions executed by 147 unique customers.

Highest Volume Category

Peak Net Liquidity — July 2023

$62,533.45

Highest Deposit Month — October 2023

$81,734.01

Lowest Liquidity Spike — April 2021

-$16,640.35

KEY INSIGHTS — PART 02

Support Operations & VIP Segmentation

Support Efficiency

288 out of 400 support cases (**72%**) relate directly to account inquiries.

Transaction Resolution Rate

88.31%

9 pending transaction cases requiring escalation.

VIP Profile: Arjun Rossi

Highest portfolio engagement client across retail & investment segments.

COMBINED VALUE

$135,095.20

INVESTMENTS BALANCE

$132,128.17

STRATEGIC ROADMAP

Actionable Business Recommendations

1. Utility Services

Enhance utility payment channels with automated auto-pay features and targeted cashback rewards.

2. Liquidity Alerts

Establish automated cash reserve alerts to mitigate seasonal deficit spikes observed in April cycles.

3. Support Optimization

Streamline transaction ticket queues to resolve 9 pending cases and boost resolution past 95%.

4. VIP Wealth Tier

Assign dedicated Relationship Managers to high-value investors like Arjun Rossi to secure retention.

EXECUTIVE VALUE

Conclusion & Decision-Making Impact

Liquidity Protection

Real-time balance surveillance across historical operational timelines.

Customer Segmentation

Identifying high-yield investment profiles for tailored engagement.

Service Optimization

Evaluating digital bill-payment preference trends for channel enhancement.

Strategic Forecasting

Data-driven forecasting to guide future banking product rollouts.



Thank You!

Questions & Discussion

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